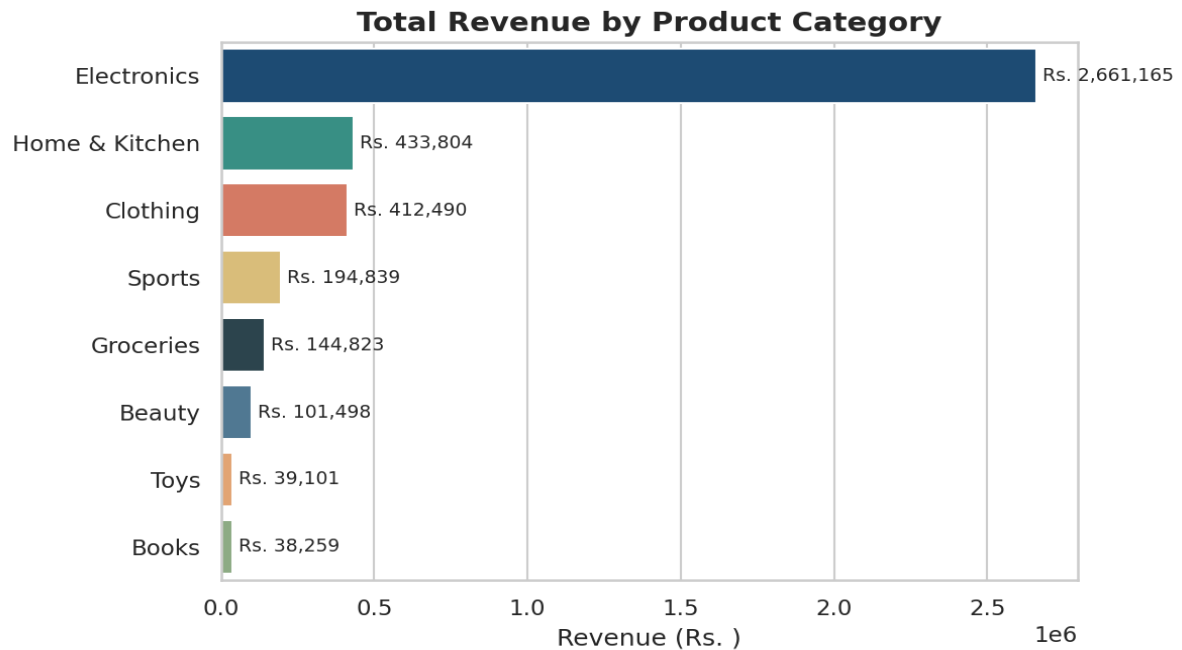


Task 3: Data Visualization Dashboard

CodSoft Data Analytics Internship

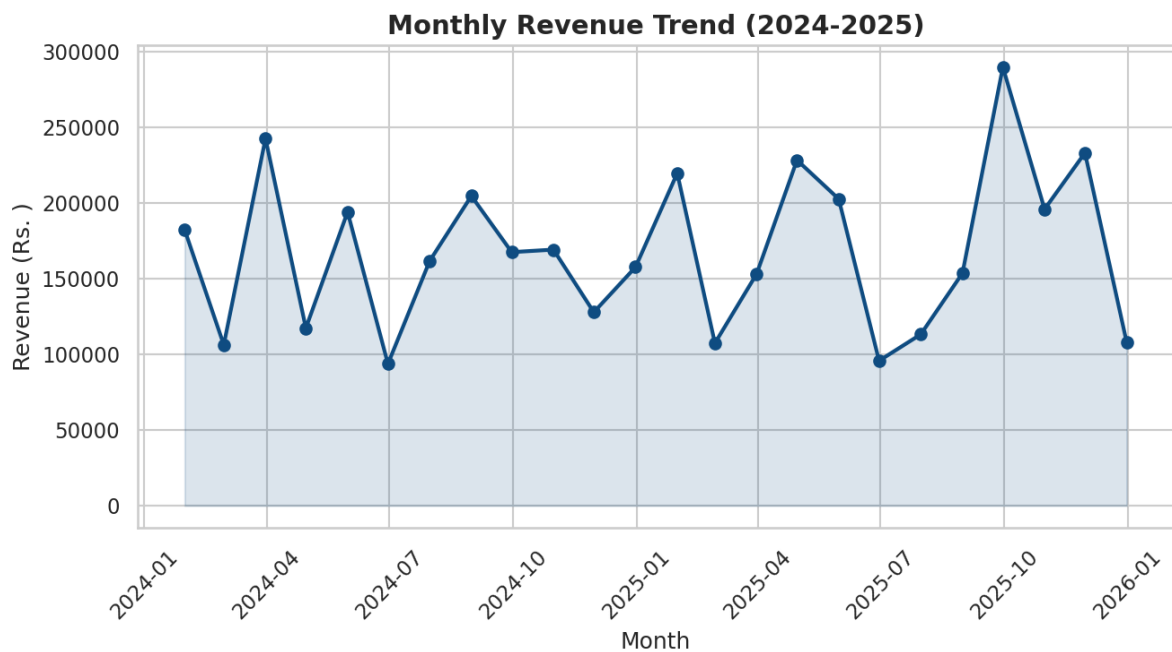
This dashboard presents key visualizations built from the customer sales dataset (1,000 orders) using Matplotlib and Seaborn, covering revenue by category, monthly trends, payment preferences, customer age distribution, and spend patterns.

1. Revenue by Product Category (Bar Chart)



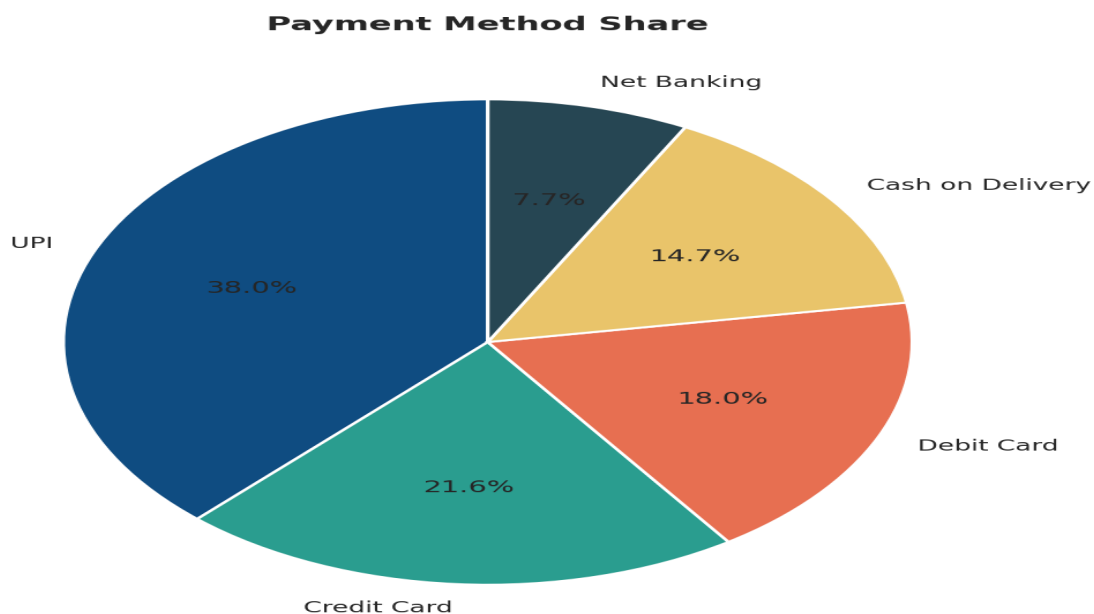
Insight: Clothing and Electronics generate the highest total revenue, followed by Home & Kitchen. Books and Toys contribute the least.

2. Monthly Revenue Trend (Line Chart)



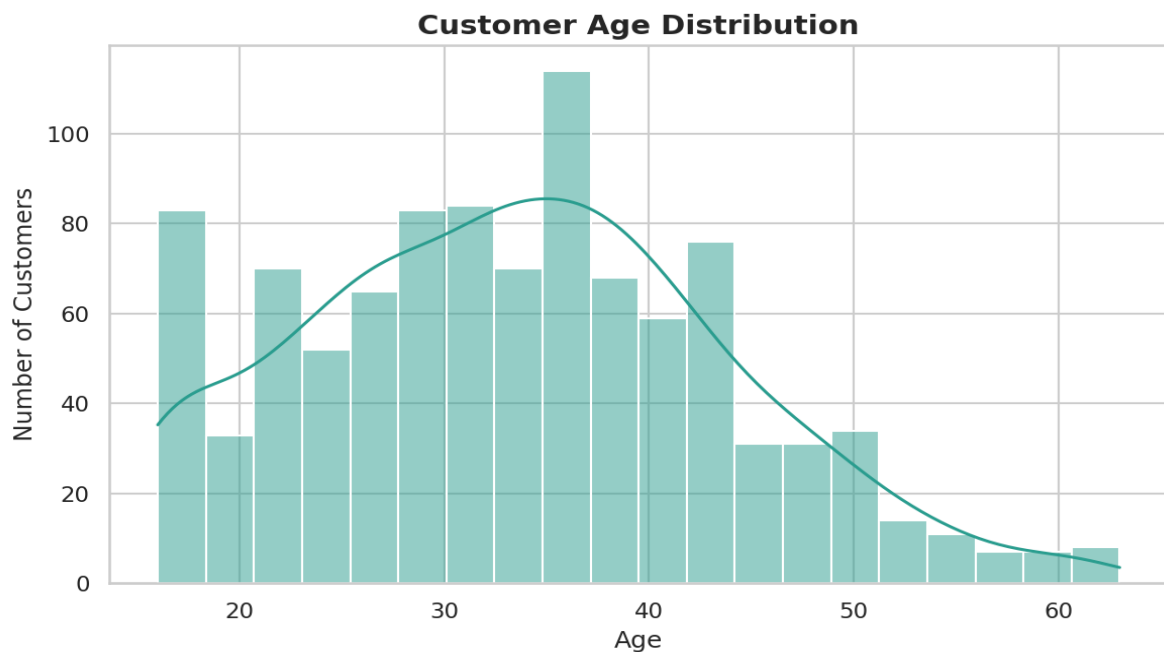
Insight: Revenue fluctuates month to month with no single dominant seasonal spike, suggesting fairly steady demand across 2024-2025.

3. Payment Method Share (Pie Chart)



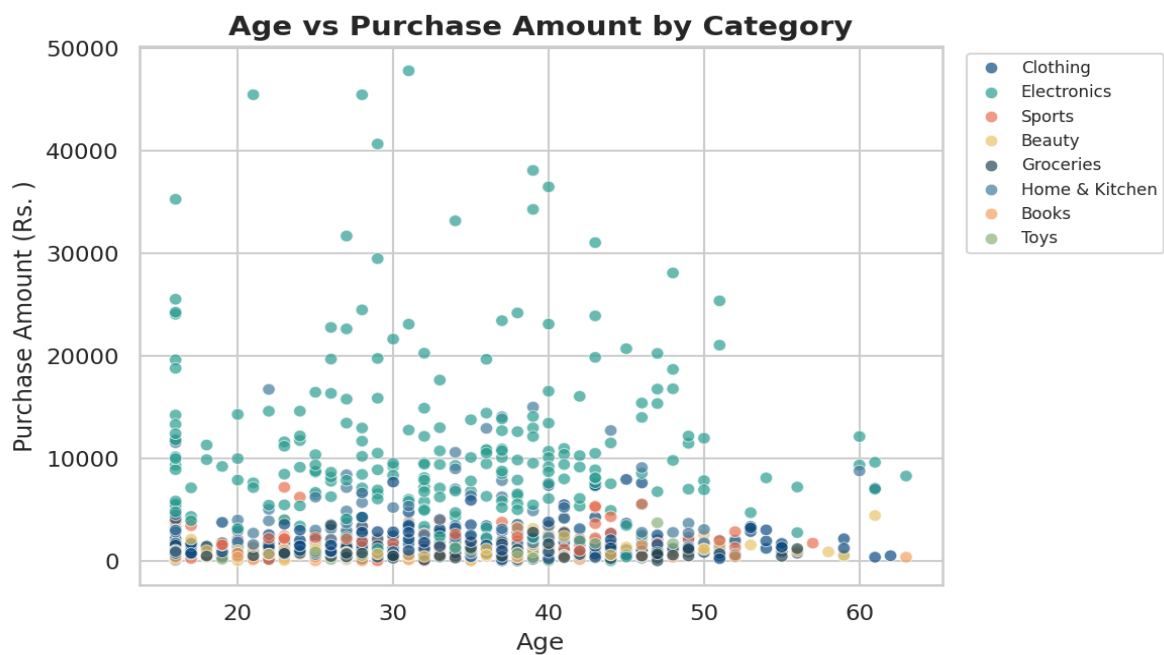
Insight: UPI is the most preferred payment method, followed by Credit Card, reflecting a strong shift toward digital payments.

4. Customer Age Distribution (Histogram)



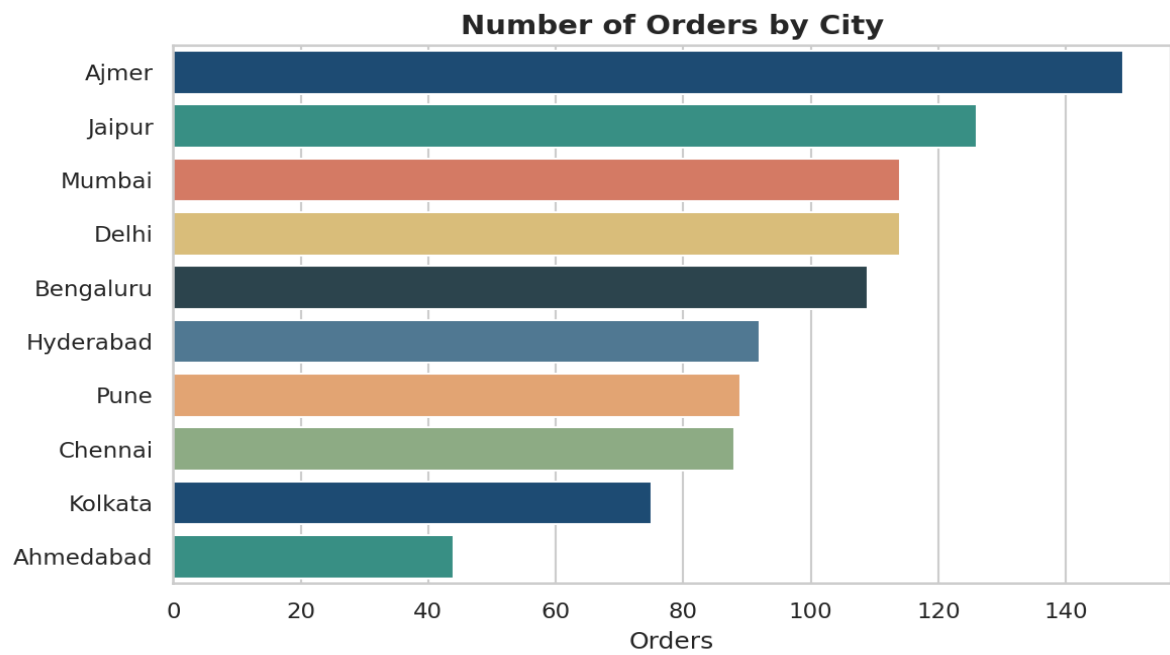
Insight: Most customers fall between 25-45 years old, with the distribution roughly bell-shaped and centered around the early 30s.

5. Age vs Purchase Amount (Scatter Plot)



Insight: No strong linear relationship exists between age and purchase amount; high-value purchases occur across all age groups, mostly in Electronics.

6. Number of Orders by City (Bar Chart)



Insight: Ajmer, Jaipur, and Delhi lead in order volume, indicating these cities are the strongest existing markets.

Bonus: Interactive Dashboard

An interactive version of this dashboard (built with Power BI / Tableau) can further allow filtering by city, category, and date range for deeper self-service exploration.